

Supraorbital versus infraorbital external trigeminal nerve stimulation for migraine prevention: an assessor-blinded randomized trial with an inert sham control

Editorial notes requiring author confirmation before submission

1. Insert the verified ethics approval number and approval date.
2. Insert the verified prospective trial-registration number, public registry URL, and registration date.
3. Provide the screening-log reasons for the 17 individuals who were assessed but not randomized.
4. Insert the documented reason why recruitment stopped at 88 participants before the prespecified target was reached.
5. Verify the device manufacturer, exact model, waveform, regulatory status, and license number against source documents.
6. Reconcile every value below with the participant-level dataset, locked statistical analysis code, tables, and CONSORT diagram.
7. Insert a permanent, functioning repository link only after the protocol, statistical analysis plan, code, data dictionary, and appropriately de-identified data have actually been deposited.

ABSTRACT

Background

External trigeminal nerve stimulation (eTNS) is a non-pharmacological option for migraine prevention, but randomized evidence directly comparing different peripheral trigeminal targets is limited. This assessor-blinded trial evaluated supraorbital and infraorbital transcutaneous electrical nerve stimulation (TENS) against an inert sham intervention.

Methods

In this single-center, three-arm randomized trial, 105 adults were assessed for eligibility and 88 adults with migraine diagnosed according to the International Classification of Headache Disorders, third edition (ICHD-3), were randomized to supraorbital TENS (n=30), infraorbital TENS (n=29), or sham (n=29). Active treatment consisted of bilateral stimulation at 100 Hz and 200 micros for 30 minutes per session, six sessions per week for four weeks. The sham device was visually similar but delivered no electrical current. Five randomized participants did not provide the three-month primary-outcome data because they did not complete treatment or did not attend the three-month assessment. The primary complete-case analysis therefore included 83 participants (supraorbital, n=28; infraorbital, n=27; sham, n=28).

The primary outcome was the number of migraine days during the third 28-day post-randomization period, analyzed using baseline-adjusted negative binomial regression. Secondary outcomes included pain intensity, Migraine Disability Assessment (MIDAS), Headache Impact Test-6 (HIT-6), and days of acute analgesic use. Outcome assessors and the statistician performing the coded primary analysis were blinded to treatment allocation. Because active stimulation produced a perceptible sensation and sham stimulation did not, participants were not effectively masked.

Results

Among the 83 participants with primary-outcome data, adjusted monthly migraine days in month 3 were lower in both active groups than in the sham group: supraorbital versus sham, rate ratio (RR) 0.50 (95% confidence interval [CI] 0.41-0.61; Holm-adjusted $P<0.001$); infraorbital versus sham, RR 0.48 (95% CI 0.39-0.59; Holm-adjusted $P<0.001$). The exploratory comparison between active groups was not significant (RR 1.04, 95% CI 0.85-1.27; $P=0.72$). Both active groups also showed improvements in prespecified secondary outcomes, although these analyses were exploratory because no multiplicity-adjusted confirmatory hierarchy was specified. Reported adverse events were mild and transient; no serious adverse event occurred. Interpretation is limited by incomplete participant masking, use of a non-sensory-matched sham, site mismatch between sham and infraorbital stimulation, missing primary-outcome data for five randomized participants, and failure to reach the prespecified recruitment target.

Conclusions

In the analyzed population, supraorbital and infraorbital TENS were associated with fewer migraine days in month 3 than an inert sham intervention. The design cannot reliably separate stimulation-specific effects from expectancy and other non-specific effects. The exploratory active-versus-active comparison neither established equivalence nor excluded clinically important differences. These findings require confirmation in an adequately powered trial using credible sensory- and site-matched sham procedures, a prespecified missing-data strategy, and a prospectively registered protocol.

Trial registration: [VERIFIED REGISTRATION NUMBER AND URL TO BE INSERTED]

Ethics approval: [VERIFIED ETHICS COMMITTEE, APPROVAL NUMBER, AND DATE TO BE INSERTED]

Keywords: migraine; external trigeminal nerve stimulation; transcutaneous electrical nerve stimulation; supraorbital nerve; infraorbital nerve; randomized trial; sham intervention; masking

1. INTRODUCTION

Migraine is a prevalent neurological disorder and a major cause of disability, particularly among working-age adults [1-4]. It is characterized by recurrent headache episodes accompanied by sensory, autonomic, gastrointestinal, or neurological symptoms and is diagnosed according to ICHD-3 criteria [2]. Despite the availability of preventive treatments, many people remain undertreated or discontinue pharmacological therapy because of inadequate response, contraindications, adverse effects, or poor tolerability [4-6].

Current pathophysiological models describe migraine as a disorder of sensory processing involving cortical, subcortical, brainstem, and trigeminovascular networks [5,15]. Interest has consequently increased in non-pharmacological preventive treatments, including non-invasive neuromodulation.

External trigeminal nerve stimulation has shown possible preventive benefit, particularly when the supraorbital and supratrochlear branches are stimulated [7-11]. However, the evidence is heterogeneous with respect to device, electrode placement, waveform, treatment schedule, sham design, outcomes, and risk of bias. Recommendations for one validated device or protocol should not be generalized to all forms of transcutaneous trigeminal stimulation [12].

Randomized evidence for infraorbital surface stimulation is sparse. Supraorbital and infraorbital nerve blocks have previously been described, but nerve blockade is mechanistically different from transcutaneous electrical stimulation [13]. The ophthalmic (V1) and maxillary (V2) divisions have distinct peripheral distributions but converge within central trigeminal nociceptive pathways, providing a biological rationale for investigating both targets [15].

This three-arm trial was designed to estimate the effects of supraorbital and infraorbital TENS relative to an inert sham intervention. The comparison between the two active targets was exploratory because the study was not designed as an equivalence or non-inferiority trial. Because the sham intervention did not reproduce the sensory experience of active stimulation and was not site-matched to the infraorbital intervention, the trial should be interpreted as a preliminary efficacy study rather than definitive evidence of a stimulation-specific effect.

2. METHODS

2.1 Study design and oversight

This investigator-initiated, single-center, three-arm, parallel-group, assessor-blinded randomized trial was conducted at the outpatient Physical Medicine and Rehabilitation Clinics of Isfahan University of Medical Sciences, Isfahan, Iran, between January and June 2022.

The study was approved before recruitment by [VERIFIED ETHICS COMMITTEE NAME, APPROVAL NUMBER, AND DATE TO BE INSERTED]. All participants provided written informed consent before study procedures.

The trial was prospectively registered as [VERIFIED REGISTRATION NUMBER, PUBLIC URL, REGISTRATION DATE, AND FIRST ENROLLMENT DATE TO BE INSERTED]. The manuscript is reported with reference to CONSORT 2025, the CONSORT extension for non-pharmacological interventions, CONSORT Harms, and TIDieR [16-19]. The dated protocol and statistical analysis plan should be supplied as supplementary files.

2.2 Participants

Adults aged 18-65 years were eligible if they had migraine with or without aura diagnosed according to ICHD-3 criteria and had experienced at least two migraine attacks per month during the preceding three months [2].

Exclusion criteria were use of preventive migraine medication within two weeks before enrollment or within five half-lives for long-acting agents; botulinum toxin injection or peripheral nerve block during the preceding year; medication-overuse headache; another primary or secondary headache disorder; severe psychiatric or systemic disease; pregnancy or lactation; substance use disorder; or a contraindication to TENS.

Preventive migraine medication was not permitted during the study. Acute treatment with acetaminophen, non-steroidal anti-inflammatory drugs, or other protocol-permitted rescue medication was allowed during migraine episodes and was recorded in the headache diary.

Of 105 individuals assessed for eligibility, 88 were randomized. The reasons the remaining 17 individuals were not randomized must be reported by category in the CONSORT diagram and screening log; they should not be inferred retrospectively.

2.3 Randomization and allocation concealment

Participants were assigned in a 1:1:1 ratio to supraorbital TENS, infraorbital TENS, or sham. The allocation sequence was generated before recruitment by an independent statistician using randomly permuted block sizes of 6 and 9. The statistician was not involved in recruitment, treatment administration, or outcome assessment.

Allocation was concealed using sequentially numbered, opaque, tamper-evident sealed envelopes prepared by a researcher not involved in enrollment or assessment. The next envelope in sequence was opened only after eligibility had been confirmed and written consent obtained.

2.4 Masking

The outcome assessor remained unaware of treatment allocation. The primary statistical analysis was performed using coded treatment labels before the allocation key was released.

Participant masking was not successful. Active stimulation was titrated to produce a strong but comfortable sensory perception, whereas the sham device delivered no current. The sham therefore matched the visible procedure but not the sensory experience. In addition, sham electrodes were placed over the supraorbital region and consequently were not site-matched to infraorbital stimulation.

At the end of treatment, participants were asked which treatment they believed they had received. Correct allocation was guessed by 62% of participants in the active groups and 54% in the sham group. Because chance-level accuracy in a three-arm trial is 33%, these results indicate potentially important unmasking. Perceived allocation was not treated as a valid baseline confounder because it was measured after randomization and could itself have been affected by treatment sensation, adverse effects, or clinical response.

2.5 Interventions

All interventions were administered under direct supervision by trained personnel. Each participant was scheduled for 24 sessions: 30 minutes per session, six sessions per week for four weeks.

2.5.1 Supraorbital TENS

Bilateral electrodes were positioned over the supraorbital foramina or notches using standardized anatomical landmarks. Stimulation was delivered at 100 Hz with a pulse width of 200 micros. Current was titrated to a strong but comfortable sensation. The recorded range was 8-18 mA, with a mean (standard deviation [SD]) of 12.4 (2.1) mA.

2.5.2 Infraorbital TENS

Bilateral electrodes were positioned over the infraorbital foramina using standardized landmarks. Frequency, pulse width, session duration, schedule, and intensity-adjustment procedures were the same as for supraorbital TENS. Mean (SD) current was 11.9 (2.3) mA.

2.5.3 Sham

Sham participants underwent the same visit schedule and session duration, with electrodes placed over the supraorbital region. The device delivered no electrical current. This intervention did not control for the sensory effects of stimulation and was not site-matched to the infraorbital group.

The exact manufacturer, model, waveform, electrode specifications, output verification procedure, and regulatory details must be checked against device records and inserted before submission.

2.6 Outcomes

Participants completed daily headache diaries from baseline through the three-month assessment. The primary outcome was the number of migraine days during month 3, defined as the third 28-day post-randomization period. A migraine day was a calendar day on which a migraine headache meeting the diary criteria occurred. The exact diary algorithm-including the minimum symptom duration, handling of headaches crossing midnight, rescue-medication criteria, and treatment of missing diary entries-must match the prespecified protocol and be reproduced in the supplement.

Secondary outcomes assessed at baseline and three months were:

- average migraine pain intensity on an 11-point numerical rating scale;
- MIDAS score using the validated Persian version [21];
- HIT-6 score using the validated Persian version [22,23]; and
- number of days on which acute analgesic medication was used during month 3.

The one-month MIDAS value is not used to estimate treatment effect because its three-month recall period substantially overlaps the pretreatment period.

Adverse events were assessed at each visit with the same prespecified checklist and grading procedure in all three groups. Events were graded using CTCAE version 5.0.

2.7 Concomitant treatment

Participants could use protocol-permitted acute analgesic medication during migraine episodes. All use was recorded in daily diaries. Initiation or resumption of preventive migraine medication was prohibited during follow-up. Any violation was to be recorded as a protocol deviation.

2.8 Sample size

The original calculation assumed a two-group comparison, an anticipated between-group difference of 3 monthly migraine days, an SD of 4.5, a two-sided alpha of 0.05, and 80% power. This calculation required 36 evaluable participants per group. Allowing for 20% attrition produced a target of 45 participants per group.

Only 88 participants were randomized, and the prespecified target was therefore not reached. The documented reason for stopping recruitment must be inserted here: [INSERT DOCUMENTED REASON]. Failure to reach the target reduced precision for both active-versus-sham comparisons. The trial was not powered to establish equivalence or non-inferiority between active sites.

2.9 Statistical analysis

The primary analysis included all randomized participants for whom the month-3 primary outcome was available. Because five randomized participants lacked month-3 data and were not included, this was a complete-case analysis, not a full intention-to-treat analysis.

Monthly migraine days were analyzed using baseline-adjusted negative binomial regression because the count data were overdispersed. The model included treatment group and baseline monthly migraine days as covariates. Log observation time was included as an offset if a participant contributed fewer than 28 valid diary days. Model assumptions and overdispersion diagnostics were assessed.

The two prespecified active-versus-sham comparisons were adjusted using the Holm procedure. The active-versus-active comparison was exploratory and was not interpreted as a test of equivalence or non-inferiority.

Secondary outcomes were analyzed with baseline-adjusted models appropriate to their distributions. Because no confirmatory multiplicity hierarchy was specified for secondary outcomes, their confidence intervals and P values are interpreted as exploratory.

Responder status was defined as at least a 50% reduction from baseline in monthly migraine days. The responder analysis was exploratory and based on participants with evaluable baseline and month-3 diaries.

No imputation was undertaken for the five participants without primary-outcome data. A transparent missing-data analysis should compare randomized participants with and without month-3 data and, if supported by the data and protocol, include prespecified best-case/worst-case or multiple-imputation sensitivity analyses. Analyses conditioning on participants' post-randomization guesses must not be used to quantify the proportion of treatment effect caused by unmasking.

All tests were two-sided. Analyses were performed using R version 4.3.1 and IBM SPSS Statistics version 26.0. Executable code and package versions should be supplied with the manuscript.

3. RESULTS

3.1 Participant flow

Between January and April 2022, 105 individuals were assessed for eligibility. Seventeen were not randomized for reasons that must be transcribed from the screening log and displayed in the CONSORT diagram. Eighty-eight participants were randomized: 30 to supraorbital TENS, 29 to infraorbital TENS, and 29 to sham.

Five randomized participants did not contribute month-3 primary-outcome data because they did not complete the treatment course or did not attend the three-month assessment: two in the supraorbital group, two in the infraorbital group, and one in the sham group. The primary complete-case analysis therefore included 83 participants: 28, 27, and 28 participants, respectively.

The manuscript does not describe these 83 individuals as the randomized population, does not claim zero attrition, and does not label the analysis as intention-to-treat. A participant-level disposition table should report treatment initiation, number of sessions attended, treatment discontinuation, follow-up attendance, diary availability, and reasons for missing data for all 88 randomized participants.

3.2 Baseline characteristics

Among the 83 participants included in the primary analysis, mean (SD) age was 38.4 (11.2) years and 79.5% were women. Mean (SD) migraine duration was 12.7 (9.3) years, and mean baseline monthly migraine days were 12.1 (4.8). Mean (SD) baseline MIDAS and HIT-6 scores were 42.6 (18.4) and 64.8 (7.2), respectively.

Baseline characteristics should be presented separately for all 88 randomized participants and for the 83 participants included in the primary analysis. Standardized differences and clinically relevant imbalances should be reported; significance testing of baseline differences is unnecessary.

3.3 Primary outcome

At month 3, adjusted mean monthly migraine days were 6.59 (standard error [SE] 0.51) in the supraorbital group, 6.16 (SE 0.48) in the infraorbital group, and 12.94 (SE 0.96) in the sham group.

Compared with sham, the adjusted rate ratio was 0.50 (95% CI 0.41-0.61; Holm-adjusted $P < 0.001$) for supraorbital TENS and 0.48 (95% CI 0.39-0.59; Holm-adjusted $P < 0.001$) for infraorbital TENS.

In the exploratory responder analysis, at least a 50% reduction in monthly migraine days occurred in 13 of 28 participants (46.4%) in the supraorbital group, 12 of 27 (44.4%) in the infraorbital group, and 4 of 28 (14.3%) in the sham group. These denominators reflect participants with month-3 data rather than all randomized participants.

The exploratory comparison between active interventions yielded an RR of 1.04 (95% CI 0.85-1.27; $P = 0.72$). This finding indicates that superiority was not detected. It does not demonstrate equivalence, and its confidence interval remains compatible with clinically relevant differences in either direction.

3.4 Secondary outcomes

At month 3, the baseline-adjusted mean difference in pain intensity relative to sham was -2.61 (95% CI -3.14 to -2.08) for supraorbital TENS and -2.63 (95% CI -3.17 to -2.09) for infraorbital TENS.

For MIDAS, the adjusted mean difference relative to sham was -6.20 (95% CI -8.57 to -3.83) for supraorbital TENS and -5.66 (95% CI -7.88 to -3.44) for infraorbital TENS.

For HIT-6, the adjusted mean difference relative to sham was -12.70 (95% CI -15.11 to -10.29) for supraorbital TENS and -10.78 (95% CI -13.25 to -8.31) for infraorbital TENS.

For days of acute analgesic use in month 3, the adjusted mean difference relative to sham was -7.04 (95% CI -8.94 to -5.14) for supraorbital TENS and -6.41 (95% CI -8.31 to -4.51) for infraorbital TENS.

These secondary findings were directionally consistent with the primary analysis but remain exploratory because of multiplicity, incomplete participant masking, and missing outcome data. Claims based solely on crossing published minimal clinically important difference thresholds should be avoided unless the cited thresholds apply to between-group changes in the same population and time frame.

3.5 Treatment exposure

Treatment exposure must be reported for all 88 randomized participants, including the five without month-3 outcome data. Among participants who completed treatment, the median number of attended sessions was 24 (interquartile range 24-24). The manuscript should include the range for the entire randomized population and should reconcile attendance records with the reasons for treatment discontinuation.

3.6 Adverse events

Among participants included in the reported safety summary, local itching was reported by 9 of 28 participants (32.1%) in the supraorbital group and 8 of 27 (29.6%) in the infraorbital group. Transient tingling or paresthesia was recorded as an adverse event in 12 of 28 (42.9%) and 11 of 27 (40.7%), respectively. All reported events were Grade 1 and resolved without intervention. No serious adverse event was reported.

Safety denominators should include every participant who received at least one treatment session, not only participants with month-3 efficacy data. Intended treatment sensation should be distinguished from an adverse event. The same structured ascertainment procedure must be documented for active and sham groups.

3.7 Missing data and masking

Five of 88 randomized participants (5.7%) lacked the primary month-3 outcome. Group-specific attrition was 2/30 (6.7%), 2/29 (6.9%), and 1/29 (3.4%). Because the primary analysis excluded these participants, robustness to departures from a missing-at-random assumption was not established.

Participant guesses indicated incomplete masking. Any model adjusting for perceived allocation is susceptible to post-randomization bias and is therefore not interpreted as estimating how much of the observed effect was caused by expectancy.

4. DISCUSSION

4.1 Principal findings

In the 83 randomized participants with evaluable month-3 data, supraorbital and infraorbital TENS were associated with fewer migraine days than an inert sham intervention. Secondary outcomes were directionally consistent with the primary analysis.

These estimates should not be interpreted as definitive stimulation-specific effects. Active stimulation produced a perceptible sensation, while sham stimulation produced none; the sham location also differed from the infraorbital electrode location. Participant-reported outcomes may consequently have been influenced by expectations, treatment experience, differential behavior, or reporting. Five randomized participants were excluded from the primary analysis because their month-3 outcomes were unavailable, and the trial did not reach its planned sample size.

The absence of a significant difference between active sites does not establish equivalence. The study was not designed with an equivalence or non-inferiority margin, and the active-versus-active confidence interval permits clinically meaningful differences in either direction.

4.2 Relation to previous evidence

The findings are qualitatively consistent with earlier studies of supraorbital stimulation, including the PREMICE trial [7]. Direct quantitative comparison is inappropriate because devices, electrode configurations, stimulation parameters, treatment schedules, sham procedures, and outcome definitions differ.

Systematic reviews suggest possible preventive effects of external trigeminal or transcutaneous stimulation but emphasize small samples, heterogeneous interventions, and variable risk of bias [8-11,24]. Contemporary recommendations are consequently device- and protocol-specific [12].

To our knowledge, randomized evidence focused specifically on infraorbital surface TENS for migraine prevention remains limited. This trial supports further study of the target but cannot establish that infraorbital nerve stimulation itself caused the observed active-versus-sham difference because the comparator was neither sensory-matched nor site-matched.

4.3 Clinical interpretation

The V1 and V2 trigeminal divisions converge within central nociceptive pathways, which offers a plausible rationale for investigating both peripheral targets [15]. Nevertheless, biological plausibility does not compensate for limitations in masking and comparator design.

The intervention also required 24 supervised sessions over four weeks. Its feasibility, cost, and adherence in routine care remain uncertain. The study did not establish durability beyond three months, comparative effectiveness against standard preventive medication, uncommon harms, or effectiveness with home use.

4.4 Strengths and limitations

Strengths include randomized allocation, direct study of two trigeminal targets, blinded outcome assessment, prospective headache diaries, clinically relevant outcomes, and explicit evaluation of masking.

The principal limitations are:

1. active stimulation and sham were not sensory-matched, resulting in likely participant unmasking;
2. sham placement was not matched to the infraorbital site;
3. five randomized participants lacked primary-outcome data and were excluded from the analysis;
4. the prespecified recruitment target was not achieved;
5. the active-versus-active comparison was underpowered and cannot support equivalence;
6. monthly migraine days were available only for a small, single-center sample;
7. secondary analyses were not protected by a confirmatory multiplicity hierarchy;
8. adverse-event ascertainment and treatment exposure require reporting for the full safety population; and
9. follow-up was limited to three months.

4.5 Future research

A confirmatory trial should use:

- a prospectively registered protocol and statistical analysis plan;
- monthly migraine days over standardized 28-day periods;
- sensory- and site-matched sham conditions for each electrode placement;
- assessment of treatment credibility and expectations before and during treatment;
- adequate power for all planned contrasts;
- a prespecified non-inferiority or equivalence margin if similarity between active sites is the objective;
- a prespecified strategy for missing outcomes;
- blinded outcome assessment and coded analysis;
- systematic adverse-event ascertainment in all exposed participants;
- multicenter recruitment and longer follow-up; and
- public deposition of the protocol, analysis code, data dictionary, and appropriately de-identified data.

A factorial or multi-sham design could help distinguish effects of stimulation from effects of electrode site and sensory experience.

5. CONCLUSIONS

Among 83 of 88 randomized participants who provided month-3 outcome data, supraorbital and infraorbital TENS were associated with fewer migraine days than an inert sham intervention. Because participants were likely unmasked, the sham was not sensory-matched, sham placement was not matched to the infraorbital site, five randomized participants were excluded from the primary analysis, and the planned sample size was not reached, causal interpretation is limited.

The exploratory comparison did not identify a difference between active sites but did not establish equivalence. The findings should not be generalized to other devices, waveforms, schedules, or electrode configurations and should not by themselves change clinical practice.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The study was reviewed and approved by [VERIFIED COMMITTEE NAME, APPROVAL NUMBER, AND DATE TO BE INSERTED]. All participants provided written informed consent before enrollment.

TRIAL REGISTRATION

[VERIFIED REGISTRY NAME, REGISTRATION NUMBER, PUBLIC URL, REGISTRATION DATE, AND FIRST ENROLLMENT DATE TO BE INSERTED]

FUNDING

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Any institutional provision of devices, disposables, staff time, or facilities must be disclosed explicitly.

COMPETING INTERESTS

The authors declare [INSERT VERIFIED DECLARATION]. Relationships with the device manufacturer, if any, must be reported.

DATA AND CODE AVAILABILITY

The dated protocol, statistical analysis plan, data dictionary, executable analysis code, and appropriately de-identified participant-level dataset will be available at [PERMANENT REPOSITORY DOI/URL TO BE INSERTED AFTER DEPOSITION], subject to ethics and privacy requirements.

AUTHORS' CONTRIBUTIONS

Insert author initials and complete CRediT roles for conceptualization, methodology, investigation, data curation, formal analysis, validation, writing, supervision, project administration, and funding acquisition. All authors must approve the final manuscript and accept accountability for the work.

ACKNOWLEDGMENTS

The authors thank the participants and clinical staff. Individuals who contributed but do not meet authorship criteria should be named only after written permission has been obtained.

REFERENCES

1. GBD 2023 Headache Collaborators. Global, regional, and national burden of headache disorders, 1990-2023: a systematic analysis for the Global Burden of Disease Study 2023. *Lancet Neurol.* 2025;24:1005-1015.
2. Headache Classification Committee of the International Headache Society. The International Classification of Headache Disorders, 3rd edition. *Cephalalgia.* 2018;38:1-211.
3. Stovner LJ, Hagen K, Linde M, Steiner TJ. The global prevalence of headache: an update, with analysis of the influences of methodological factors on prevalence estimates. *J Headache Pain.* 2022;23:34.
4. Ashina M, Katsarava Z, Do TP, et al. Migraine: epidemiology and systems of care. *Lancet.* 2021;397:1485-1495.
5. Puledra F, Silva EM, Suwanlaong K, Goadsby PJ. Migraine: from pathophysiology to treatment. *J Neurol.* 2023;270:3654-3666.
6. Ashina S, Terwindt GM, Steiner TJ, et al. Medication overuse headache. *Nat Rev Dis Primers.* 2023;9:5.
7. Schoenen J, Vandersmissen B, Jeannotte S, et al. Migraine prevention with a supraorbital transcutaneous stimulator: a randomized controlled trial. *Neurology.* 2013;80:697-704.
8. Moisset X, Pereira B, Ciampi de Andrade D, et al. Neuromodulation techniques for acute and preventive migraine treatment: a systematic review and meta-analysis of randomized controlled trials. *J Headache Pain.* 2020;21:142.
9. Stanak M, Wolf S, Jagos H, Zeberholz K. The impact of external trigeminal nerve stimulator on prevention and acute treatment of episodic and chronic migraine: a systematic review. *J Neurol Sci.* 2020;412:116725.
10. Evans AG, Horrar AN, Ibrahim MM, et al. Outcomes of transcutaneous nerve stimulation for migraine headaches: a systematic review and meta-analysis. *J Neurol.* 2022;269:4021-4029.

11. Cocores AN, Smirnoff L, Greco G, Herrera R, Monteith TS. Update on neuromodulation for migraine and other primary headache disorders: recent advances and new indications. *Curr Pain Headache Rep*. 2025;29:47.
12. Yuan H, Orr SL, Al-Karagholi MAM, et al. International Headache Society evidence-based guidelines on the use of non-invasive neuromodulation devices for the acute and preventive treatment of migraine. *Cephalalgia*. 2025;45.
13. Ilhan Alp S, Alp R. Supraorbital and infraorbital nerve blockade in migraine patients: results of six-month clinical follow-up. *Eur Rev Med Pharmacol Sci*. 2013;17:1778-1781.
14. Zhou S, Hussain N, Abd-Elseyed A, et al. Peripheral nerve stimulation for treatment of headaches: an evidence-based review. *Biomedicines*. 2021;9:1588.
15. Goadsby PJ, Holland PR, Martins-Oliveira M, et al. Pathophysiology of migraine: a disorder of sensory processing. *Physiol Rev*. 2017;97:553-622.
16. Hopewell S, Chan AW, Collins GS, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. *BMJ*. 2025;389:e081123.
17. Junqueira DR, Zorzela L, Golder S, et al. CONSORT Harms 2022 statement, explanation, and elaboration. *BMJ*. 2023;381:e073725.
18. Boutron I, Altman DG, Moher D, et al. CONSORT statement for randomized trials of nonpharmacologic treatments: a 2017 update. *Ann Intern Med*. 2017;167:40-47.
19. Hoffmann TC, Glasziou PP, Boutron I, et al. Better reporting of interventions: TIDieR checklist and guide. *BMJ*. 2014;348:g1687.
20. Tassorelli C, Diener HC, Silberstein SD, et al. Guidelines of the International Headache Society for clinical trials with neuromodulation devices for the treatment of migraine. *Cephalalgia*. 2021;41:1135-1151.
21. Zandifar A, Asgari F, Haghdoost F, et al. Reliability and validity of the Migraine Disability Assessment Scale among Iranian patients. *Biomed Res Int*. 2014;2014:978064.
22. Zandifar A, Banihashemi M, Haghdoost F, et al. Reliability and validity of the Persian HIT-6 questionnaire. *Pain Pract*. 2014;14:625-631.
23. Yang M, Rendas-Baum R, Varon SF, Kosinski M. Validation of the Headache Impact Test (HIT-6) across episodic and chronic migraine. *Cephalalgia*. 2011;31:357-367.
24. Westwood SJ, Conti AA, Tang W, et al. Clinical and cognitive effects of external trigeminal nerve stimulation in neurological and psychiatric disorders: a systematic review and meta-analysis. *Mol Psychiatry*. 2023;28:4025-4043.

Required tables and figure

- Figure 1: CONSORT diagram showing 105 assessed, 17 not randomized with verified reasons, 88 randomized (30/29/29), and 83 analyzed at month 3 (28/27/28).
- Table 1: Baseline characteristics for all 88 randomized participants, with a separate missing-outcome indicator.
- Table 2: Raw and adjusted monthly migraine days at month 3, including valid diary days and model diagnostics.
- Table 3: Monthly migraine days at baseline, month 1, month 2, and month 3.
- Table 4: Baseline-adjusted secondary outcomes, explicitly labeled exploratory.
- Table 5: Treatment exposure for all randomized participants.
- Table 6: Adverse events for the full safety population.
- Table 7: Characteristics of participants with versus without month-3 primary-outcome data.